

20/3/26

DSA - Assignment-2

Inor INORDER-TRAVERSAL (IT)

X = T.root

while x != Nil

if x.Left == Nil

print(x.Key)

x = x.right

else:

pre = x.Right

~~while pre != Nil and pre !=~~

while pre.right != Nil and pre.right == x:

pre = pre.right

if pre.right == Nil:

pre.right = x

else:

pre.right = Nil

print(x.Key)

x = x.right

Wh? because Once
it breaks the link
it needs to print Key.

Q2

Given:- professor Kilmer claims, that there are three sets

Set A: the Key to the left of Search path

Set B: the Keys on the Search path

Set C: the Keys to the right of the Search path.

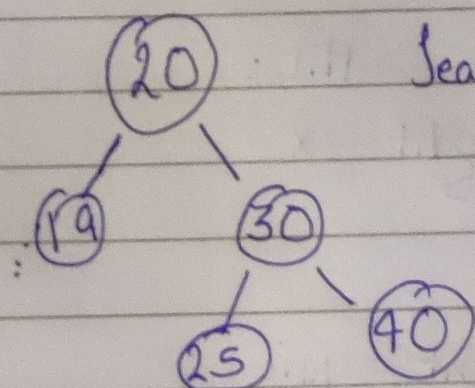
→ he claims that any three Keys

$a \in A, b \in B, c \in C$ must satisfy

$$a \leq b \leq c$$

Solution:- Professor's claim can be counter by a simple example

(1)



Search for Key = 40
all Professor

$$A = \{25, 19\}$$

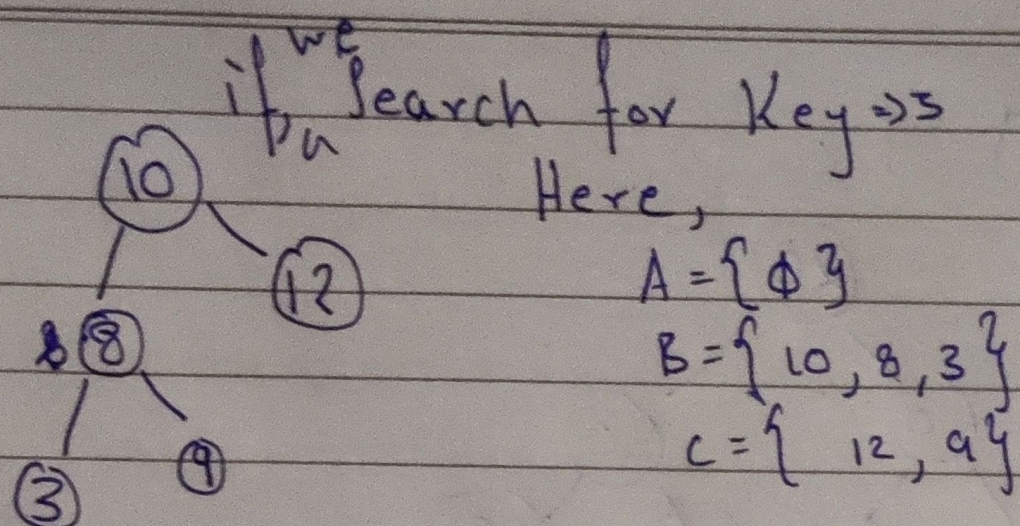
$$B = \{20, 30, 40\}$$

$$C = \{\emptyset\}$$

→ if you take $\{20\} \in B, \{25\} \in A$

then,

$25 \not\leq 20$, thus Contradicts professor's claim.



$\rightarrow$ Claim $\Rightarrow a \leq b \leq c$
 $\rightarrow 10 \leq \{10\} \in B, \{9\} \in C$

$10 \not\leq 9$
 thus $b \not\leq c$

Hence this Contradicts Professors claim

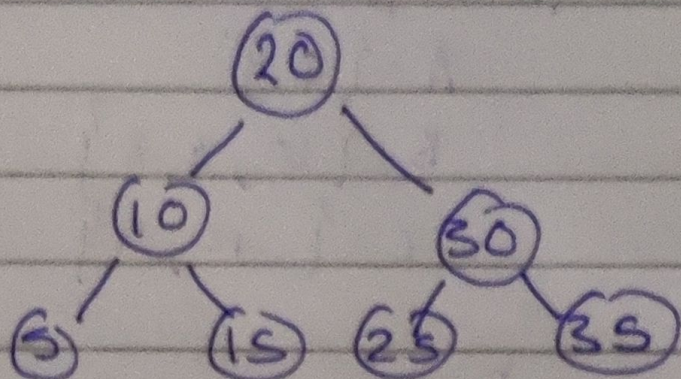
for Correctness $\rightarrow$

By Search Path we mean that the Keys that lie on the path leading to Key we are looking for

By Keys on the left of path that ~~we~~ ^{it} means that all the Keys to the left of Search path the entire left sub tree as well

By Keys on the Right of the Search path it ~~is~~ means that the Keys on the right of path we are looking for all the entire right sub tree.

To Explain clearly ,



Search {30}

$A = \{ 10, 5, 15, 25 \}$

$B = \{ 20, 30 \}$

$C = \{ 35 \}$

Q4]

The pseudocode for Tree-predecessor is as follows:-

Node pointer $\leftarrow$

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TREE-PREDECESSOR( $x$ )
  if ( $x.left \neq NIL$ )
    return Maximum( $x.left$ )
  else
     $y = x.p$ 
    while ( $y \neq NIL$  and  $y = x.left$ )
       $x = y$ 
       $y = y.p$ 
    if ( $y \neq NIL$ )
      print( $y.Key$ )
    else
      print(NULL)
  
```